

SUKKUR IBA UNIVERSITY
DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

FYDP Progress Report

Project Title:
SafeDrive AI: Intelligent dashboard device for driver anomaly detection

Muhammad Usman, 133-22-0016

Asma Channa, 133-22-0042

Abdul Moiz, 133-22-0032

Kashish Thorani, 133-22-0015

Supervisor

Dr. Abdul Sattar Chan

Sukkur IBA University

Supervisor

Engr. Umair Ayaz Kamangar

Sukkur IBA University

Progress Report Submission Date:

Details of FYDP Work:

Final Year Design Project Work	Hardware Based (%)	Software Based (%)
	60%	40%

	In term of (%)	Description
Executed Work	70%	• Applied foundational data augmentation techniques to increase dataset variability and effectively address class imbalance. • Trained the YOLOv8n-cls model using multiple hyperparameter configurations, including variations in batch size, learning rate, and training epochs, to optimize classification performance. • Observed that the YOLOv8-cls model delivered reliable and higher-accuracy results based on the conducted experiments. • Completed the full environment setup on the Raspberry Pi for on-device inference. • Deployed and tested the trained YOLOv8-cls model on the Raspberry Pi using a sample video to validate operational performance. • Introduced an additional target class, “Drowsy” to the existing dataset, extending the current class taxonomy. • Used Gamma Correction to darken mid-tones and crush shadows creating a high-contrast looks of low-light environments. • Also use channel-specific scaling and desaturation to mimic conditions where blue tones dominate as the human eye loses color sensitivity. • Retrained the YOLOv8-cls model using the expanded dataset and validate the updated performance metrics. • Converted the retrained model to .onnx format to enable deployment on the Raspberry Pi. • Tested the model in real-time environment in car and observed the behavior of the system.
Remaining Work	30%	• Perform model quantization to convert the AI from 32-bit to 8-bit precision. • Physically mount and connect the Camera Module to the Raspberry Pi for live video input. • Wire the physical speaker to the Raspberry Pi to enable real-world alerts. • Configure the entire software stack as a Linux System Service. • Ensure the device starts monitoring automatically as soon as the car is turned on. • Conduct final "in-vehicle" testing to calibrate the camera under real sunlight and night-driving conditions. • Fine-tune the alert sensitivity to ensure the system is helpful without being annoying to the driver.

Comments by Supervisor/Co-Supervisor:

Is the FYDP Project/Thesis ready for Evaluation I/II/III/final presentation? No ☐ Yes ☐

Signature

Supervisor
